



Dairy & Milk Cooperative Management System

Team ID: 6 | Section: S9 | Members: S. Prem Sai – 2520030561, V. Nitish Chowdhary – 2520030179, T. Jaswanth Reddy – 2520030284 | Guide: Dr. Y Prasanthi

Database Systems Engineering and Distributed Backend Development (25CS1302E) | KLH Bachupally Campus, CSE | PBL 2026-27

Abstract

- Web platform for dairy parlour merchants and cooperative admins to manage products, stock, POS billing, payments and invoices — built with React, Node.js microservices and MySQL.
- Outcome:** live inventory on every sale, plus automatic stock and payment fraud flags.

Problem Statement & Objectives

- Problem:** Paper registers let stock drift from sales and hide cash leakage until manual tallying.
- Objective 1:** Atomic multi-item sales — reserve stock, verify payment, commit in one transaction.
- Objective 2:** Cash and Cashfree UPI payments, server-side pricing.
- Objective 3:** Automatic stock & payment discrepancy flags.
- Scope:** Admin/Owner & Merchant roles; stock, sales, payments, reports.

Existing vs Proposed System

Feature	Existing System	Proposed System
Stock	Manual register	Auto-updated per sale
Billing	Handwritten bills	Multi-item POS + invoice
Payments	Cash, unrecorded	Cash + UPI ledger
Revenue	End-of-day tally	Live admin dashboard
Gap	Leakage unnoticed	Automatic fraud flags

Feasibility Study

- Technical:** Team skilled in JavaScript, React, Node.js and SQL; the whole stack runs on a standard laptop.
- Economic:** Fully open-source stack (React, Express, MySQL); Cashfree sandbox and free-tier hosting cost nothing to develop and demo.
- Operational:** Role-based dashboards with simple forms — merchants bill and record stock with minimal training; admins review flags in one screen.
- Schedule:** Built in phases — database and core modules first, then microservices and the payment saga — within the 2026-27 trimester.

Results & Testing

- 9 integration test suites cover saga rollback & recovery, gateway cutover, reporting and service restart.
- Seed data correctly raises a ₹700 payment flag and a 10-unit stock flag.

Conclusion

- Delivered a working role-based dairy POS with inventory, payments and invoicing.
- Meets all three objectives: atomic sales, cash + UPI, automatic discrepancy flags.

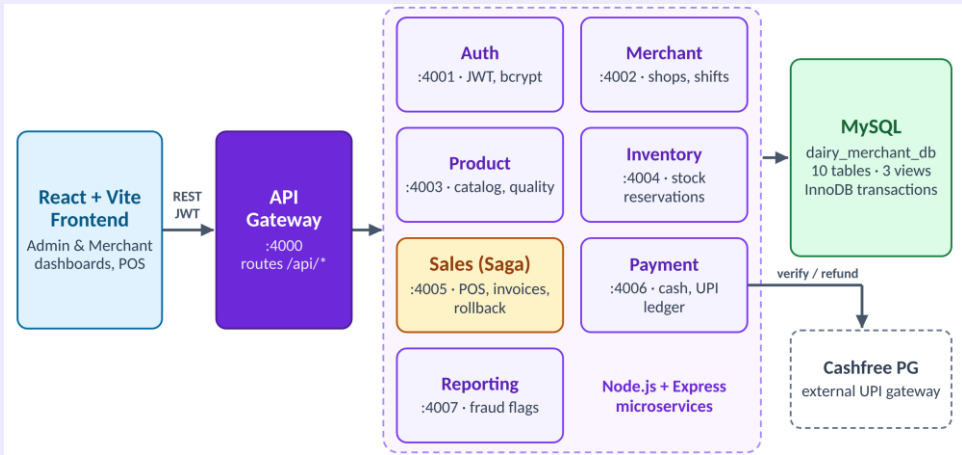
Future Scope

- Farmer milk collection with Fat/SNF-based rate calculation and payment statements.
- Email/SMS alerts, Docker deployment and a CI/CD pipeline.

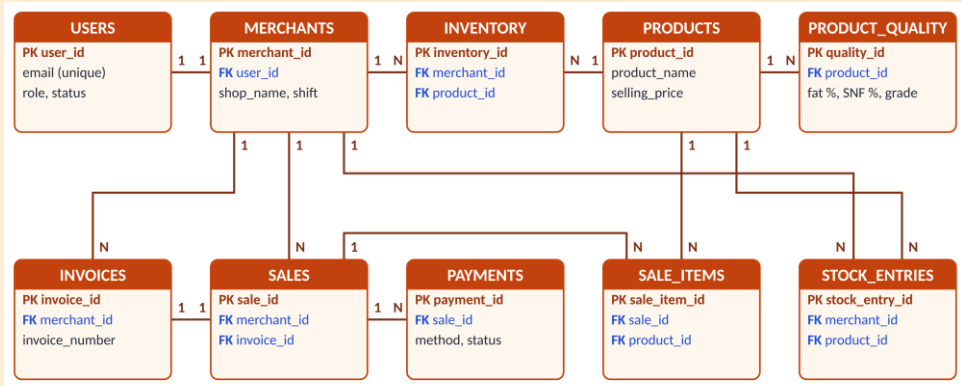
Tools & Technologies

- Front-End:** React 18 (Vite), React Router, Axios
- Back-End:** Node.js + Express microservices behind an API Gateway; REST APIs; JWT + bcrypt auth
- Database:** MySQL (InnoDB) — 10 tables, 3 views, indexes, ACID transactions
- Tools:** Git & GitHub, Cashfree PG (sandbox), Postman, VS Code, Draw.io

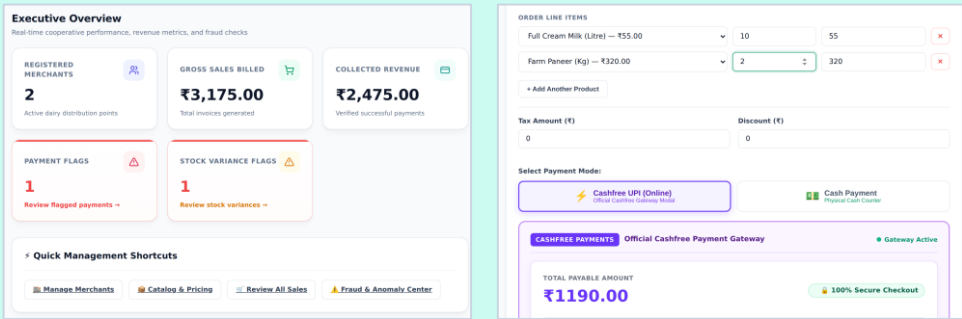
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Admin dashboard & fraud flags

Merchant POS with UPI billing

